

INFO 298: Directed Group Study

Causal Detection of Contamination Laundering in LLM Post-Training

Start Date: Fall 2026

Format: Weekly seminar + independent research

Target Audience: ML/AI-focused MIMS students pursuing research or research-engineer roles

Grading: S/U

Units: 3

Students:

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Background & Goal

Open reasoning-model post-training (s1, Tulu 3, OpenThoughts, and their successors) reports large benchmark gains, but those gains are only as trustworthy as the decontamination pipelines meant to guarantee no train/test overlap. Our prior work (ANLP Sp26, “Are Reasoning Model Benchmark Gains Real?”) found that decontamination filters at all three major open post-training projects let real leakage through, and that a difference-in-differences perturbation framework can detect a behavioral fingerprint of contamination — shorter, less self-correcting chain-of-thought traces — even when the accuracy-based causal estimate is underpowered.

The wider literature converges on two hard, unresolved problems that this project directly targets:

1. **Membership-based detection is fragile and evadable.** Black-box membership inference sits at $AUC \approx 0.5$ (chance) in rigorous comparisons, and paraphrase-laundered contamination — rewording a benchmark item before training on it — has been shown to defeat n-gram decontamination entirely, letting a fine-tuned 13B model match GPT-4 on benchmarks it never should have seen cleanly.
2. **Production decontamination pipelines are still primitive.** Even recent open-weight releases (Apertus, OLMo 3, Meditron) rely on n-gram/token-overlap matching with light semantic filtering bolted on — the exact class of method known to be evadable.

Core hypothesis: membership-based detectors (n-gram, embedding similarity, Min-K%, LLM-as-judge) degrade toward chance as contamination is increasingly laundered through paraphrase, but causal *behavioral* detection — measuring whether a model’s problem-solving shows the shortcut-reliance fingerprint under controlled perturbation — continues to work, because it measures the downstream capability effect rather than the upstream string match. A launderer can hide the words; they cannot as easily hide the shortcut.

This project runs a controlled contamination-injection study — rather than an observational audit — to establish ground truth, sweep laundering strength as an independent variable, and produce a clean, citable headline result: an evasion curve where membership-detection AUC collapses while causal behavioral-detection AUC holds. The semester’s work is scoped to produce a submission-ready draft for ICML 2027 (deadline Jan 22, 2027), with ICLR 2028 as a fallback venue if additional experimental runway is needed.

Learning Objectives

- Design and execute a controlled causal experiment (contamination injection + graded paraphrase laundering + LoRA fine-tuning) that isolates the causal effect of contamination from confounds that plague observational audits.
 - Implement and benchmark the current state-of-the-art contamination detection methods (n-gram overlap, dense embedding retrieval, Min-K%, LLM-as-judge) and quantify their degradation under adversarial laundering.
 - Extend a difference-in-differences behavioral detection framework (perturbation testing, chain-of-thought fingerprinting) and evaluate its robustness against the same laundering sweep.
 - Produce a reusable, released research artifact (injection pipeline, labeled checkpoints, detector suite) suitable for a Datasets & Benchmarks track submission as a fallback to the main experimental contribution.
 - Practice end-to-end ML research craft: literature positioning, experimental design under resource constraints, statistical power planning, and technical writing for a top-tier ML venue.
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Deliverables

- **Mid-semester presentation:** a demo of the laundering-injection pipeline and preliminary detection results across at least one laundering condition, with a clear statement of what has and hasn’t yet been validated.

- **Final deliverable:** a submission-ready research paper draft targeting ICML 2027, including the full evasion-curve result (membership-detection AUC vs. laundering strength, overlaid against behavioral-detection AUC), the released injection/detection toolkit with documentation, and a README describing the experimental pipeline, model checkpoints used, and reproduction steps.
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Course Materials

Note: this list is subject to change as new work emerges during the semester — contamination/evaluation is a fast-moving subfield.

Foundational Papers

- Sainz et al. (2023), *NLP Evaluation in Trouble: On the Need to Measure LLM Data Contamination for Each Benchmark* (EMNLP Findings).
- Carlini et al. (2021), *Extracting Training Data from Large Language Models* (USENIX Security).
- Yang et al. (2023), *Rethinking Benchmark and Contamination for Language Models with Rephrased Samples* — the foundational paraphrase-evasion result.
- Deng et al. (2024), *Investigating Data Contamination in Modern Benchmarks for Large Language Models*.

Post-Training / RL-Era Contamination (our project's frontier)

- Kocyigit & Yildirim (2026), *The Impact of Post-Training on Data Contamination*.
- Wu et al. (2026), *Reasoning or Memorization? Unreliable Results of Reinforcement Learning Due to Data Contamination* (AAAI).
- *Detecting Data Contamination from Reinforcement Learning Post-training* (Self-Critique method, 2026).
- *LaRA: Layer-wise Representation Analysis for Detecting Data Contamination in RL Post-Training* (2026).

Detection Fragility & Evasion (the problem we're targeting)

- *Evading Data Contamination Detection for Language Models is (too) Easy* — the paper this project directly rebuts/extends.
- Shi et al. (2023), *Detecting Pretraining Data from Large Language Models* (Min-K%).
- Sun et al. (2025), *The Emperor's New Clothes in Benchmarking? A Rigorous Examination of Mitigation Strategies for LLM Benchmark Data Contamination*.
- Nourbakhsh et al. (2026), *Are LLM Benchmarks Already Contaminated? A Systematic Review of Contamination Detection Methods* (GEM Workshop) — field-level survey identifying RL/post-training auditing as immature.

Dynamic Benchmarks & Causal Framing

- White et al. (2024), *LiveBench: A Challenging, Contamination-Free LLM Benchmark*.
- ConTAM (2025) — causal contamination analysis via controlled fine-tuning injection; closest prior work to our experimental design.
- GSM-Plus / MATH-Perturb / GSM-SEM — perturbation-based robustness benchmarks informing our laundering and perturbation design choices.

Datasets & Tooling

- s1K, Tulu 3 SFT mixture, OpenThoughts-114K (released, post-decontamination training sets — baseline audit targets from our prior work).
 - MATH-500, and a code-domain benchmark (HumanEval / LiveCodeBench-style) for the cross-domain extension.
 - HuggingFace transformers / peft (LoRA fine-tuning), FAISS (dense retrieval baselines).
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Research Phases

The semester is organized into six rolling phases rather than a fixed weekly schedule, since experimental research timelines shift as results come in. Phases are expected to overlap somewhat, particularly toward the middle of the semester.

Phase 1: Ground Truth Design & Literature Positioning

Early semester

Objective: Establish a verified clean problem pool and finalize the experimental design against prior work.

- Curate and verify an uncontaminated problem set (math primary; code as stretch), cross-checked against known training corpora.
- Formally differentiate this project’s controlled-injection design from ConTAM and our own prior observational audit.
- Define the laundering-strength conditions (verbatim / light paraphrase / heavy paraphrase / numeric-perturbation-laundered) as the study’s independent variable.
- Specify the membership-detection baseline suite and finalize the causal DiD design at adequate statistical power ($N \geq 5$ samples/prompt, majority vote) — directly addressing the power limitation flagged in our prior paper.

Phase 2: Controlled Contamination Injection Pipeline

Early-to-mid semester

Objective: Build and validate the graded-launders generation pipeline.

- Implement paraphrase generation at each laundering strength via structured LLM prompting, with confidence/quality validation.
- Implement the numeric-perturbation-launders condition, reusing and extending our existing perturbation infrastructure.
- Validate laundering quality: confirm paraphrases preserve the answer and underlying reasoning structure.
- Construct per-condition training data splits and select base model(s) for LoRA fine-tuning.

Phase 3: Fine-Tuning Under Graded Laundering

Mid semester

Objective: Execute and validate LoRA fine-tuning across laundering conditions.

- Run SFT fine-tuning for each laundering-strength condition plus clean control.
- Confirm ground-truth contamination status is correctly reflected in each checkpoint (sanity-check via known-verbatim condition).
- Freeze checkpoints and produce the labeled model set for the detection phase.
- **Mid-semester presentation** falls within this phase.

Phase 4: Membership-Based Detection Baselines

Mid-to-late semester

Objective: Run the standard detector suite against all conditions and characterize its degradation.

- Apply n-gram overlap, dense embedding retrieval, Min-K%, and LLM-as-judge verification across all laundering conditions.
- Compute detection AUC per method per laundering strength.
- Identify the laundering threshold at which each method collapses toward chance — the core “evasion” result.

Phase 5: Causal Behavioral Detection & Validation

Late semester, overlapping with Phase 4

Objective: Apply the DiD/perturbation framework to all injected checkpoints and extract the behavioral fingerprint.

- Run number-swap and surface-noise perturbations across all conditions at adequate power.
- Compute DiD estimates and bootstrap confidence intervals per laundering condition.

- Compute CoT features (self-correction rate, null rate, hedging, math-token density) per condition.
- Test whether behavioral-detection AUC holds steady across laundering strength where membership detection degrades — the paper’s central result.

Phase 6: Synthesis, Writing, and Release

End of semester, extending into early January

Objective: Consolidate results, write up the paper, and package the release.

- Produce the overlaid evasion-curve figure (membership AUC vs. behavioral AUC, by laundering strength).
- Run stretch-goal ablation if time allows: a reduced-scope GRPO/RL post-training condition to probe SFT-vs-RL differences.
- Write up methodology, results, and related-work positioning against the evasion and post-training-contamination literature.
- Package and document the release: injection pipeline, labeled checkpoints, detector suite, and reproduction instructions.
- Continue revision and internal review into early January ahead of the ICML 2027 submission deadline (Jan 22, 2027), which falls just after the semester’s formal end.